

Highlights

2D Image Classification with Random Convolutional Kernels: A ROCKET-Based Approach

Felicitas Lock, Christopher Bonenberger

- Adapts ROCKET's random-kernel, statistics-based paradigm from 1D time series to 2D images.
- Systematic ablations isolate the effect of feature statistics, 1D vs. 2D kernels, padding, preprocessing, and kernel budget.
- Final 2D-ROCKET reaches 99.41% (MNIST) / 73.43% (CIFAR-10) without any gradient-based training.
- Trains in seconds to minutes, versus hours for a linear SVM baseline, while beating it decisively in accuracy.
- Slightly beats a deliberately shallow CNN baseline, but as a flat, single-stage method it has no mechanism for the multi-scale composition of deeper CNNs; proposed instead for texture-dominated imagery, where a KTH-TIPS proof-of-concept reaches 94.7%.
- A ten-seed, paired-significance real-world evaluation (NEU, DTD, Chest X-Ray) tests the texture-domain scoping directly: 2D-ROCKET wins significantly on texture-dominated NEU/DTD and loses significantly, more narrowly, on non-texture Chest X-Ray.
- Adding a previously unimplemented positional-pooling feature (LSPV) closes the Chest X-Ray gap to non-significance ($p = 0.22$) at no runtime cost, confirming the mechanistic explanation for that loss.
- A freshly implemented 1D-ROCKET applied to real Sentinel-1 raw radar time series for RFI detection beats a linear SVM but not a CNN on this small-sample (216-example) task, extending the texture-vs-localized-anomaly account to a non-image domain.

2D Image Classification with Random Convolutional Kernels: A ROCKET-Based Approach

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ABSTRACT

Deep learning achieves strong performance in image classification but often requires costly training and substantial computational resources. This paper explores a lightweight alternative by adapting the ROCKET framework from time-series classification to two-dimensional image data. The proposed 2D-ROCKET approach applies large banks of random 2D convolutional kernels and summarizes their responses using simple aggregation statistics, enabling efficient feature extraction without learned filters and, structurally, without the stacked, hierarchical multi-scale composition that gives deep CNNs their representational power. MNIST and CIFAR-10 are used here not as target applications but as controlled, well-understood benchmarks to probe the method's ceiling: preserving spatial structure with 2D kernels improves performance compared to flattened 1D variants, and a final configuration (5,000 mixed-scale kernels, moderate dilation, valid padding) reaches 99.41% accuracy on MNIST and 73.43% on CIFAR-10, slightly ahead of a deliberately shallow, two-layer CNN baseline on both benchmarks and far ahead of a linear SVM, at a fraction of the SVM's training time. This should not be read as evidence that 2D-ROCKET rivals deep CNNs in general: the shallow baseline itself has little of the multi-scale hierarchical depth that distinguishes modern, deeper architectures, which were out of scope for this comparison. Object recognition on natural images is, in any case, not the domain 2D-ROCKET is proposed for: a proof-of-concept on the texture-dominated KTH-TIPS dataset (94.7% accuracy, versus 43.5%/52.4% for SVM/CNN) indicates that the method's real niche is texture-like, repetitive-pattern imagery – material and surface textures, inspection data, or structured surface signals – where classification does not depend on the multi-scale, compositional object structure that only deep, hierarchical architectures can capture. This texture-domain scoping is tested directly on three real-world datasets – industrial surface-defect inspection (NEU), a texture-attribute benchmark (DTD), and chest X-ray pneumonia screening – over ten seeds with paired significance testing. 2D-ROCKET wins significantly on the texture-dominated datasets (NEU, DTD) and loses significantly, by a smaller margin, on the non-texture, localized-pathology chest X-ray task; adding a previously unimplemented positional-pooling feature (LSPV) closes the chest X-ray gap to statistical indistinguishability at negligible cost, with a capacity-matched control attributing roughly three-quarters of that gain to positional resolution rather than added feature capacity. A further test on genuine 1D time series – a freshly implemented 1D-ROCKET applied to raw Sentinel-1 SAR echo lines for radio-frequency-interference detection (216 examples) – beats a linear SVM but not a CNN, consistent with a texture-vs-localized-anomaly distinction rather than sample size alone determining where ROCKET-family methods have an edge. Overall, 2D-ROCKET provides a simple, training-efficient, and interpretable baseline, and highlights random convolutional feature extraction as a promising direction for lightweight, texture-oriented image classification, rather than as a competitor to deep CNNs on general object recognition.

1. Introduction

Image classification has become a central task in machine learning, with deep learning models achieving remarkable accuracy across a wide range of datasets. Classic machine learning algorithms, both on images and on time series, often require expensive feature-engineering steps, resulting in a demand for high computing effort and correspondingly powerful hardware (Ismail Fawaz, Forestier, Weber, Idoumghar and Muller, 2019). Moreover, these models often come with large memory requirements and limited interpretability. Exploring alternative approaches that are computationally efficient, lightweight, and conceptually simple is therefore of growing interest.

RandOm Convolutional KERNel Transform (ROCKET) is a method originally developed for time-series classification that offers high efficiency through large banks of random convolutional kernels combined with simple aggregation

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features and a linear classifier (Dempster, Petitjean and Webb, 2020a). Investigating whether this idea can be transferred to image data may yield insights into efficient, non-deep alternatives for image classification.

The main objective of this work is to investigate whether ROCKET-style feature extraction can be effectively adapted to image classification. Concretely, we (i) design a ROCKET-inspired 2D feature-extraction pipeline for images using random convolutional kernels and simple aggregation statistics, (ii) analyze the impact of kernel size, dilation, kernel count, padding, and feature selection through controlled ablations, (iii) evaluate the resulting pipeline, termed 2D-ROCKET, on MNIST and CIFAR-10, (iv) compare it against a minimal 1D-ROCKET baseline, a linear SVM, and a lightweight CNN, and (v) test the texture-domain scope this comparison points toward against a lightweight CNN on three independent, real-world datasets under a properly powered, multi-seed statistical protocol, rather than resting that scoping claim on single-run comparisons alone.

The scope is deliberately restricted: all convolutional kernels remain fixed after random initialization, hyperparameter search is limited to small predefined grids, and comparisons are made against lightweight rather than state-of-the-art baselines. The goal is not to outperform modern deep networks, but to isolate the effect of spatially structured random convolutions and quantify the resulting accuracy/cost trade-offs.

A related but separate scoping point concerns the target application. MNIST and CIFAR-10 are used here because they are standard, well-characterized benchmarks that make the ablations reproducible and comparable to prior work – not because natural-object recognition is where 2D-ROCKET is expected to be competitive. A deep CNN builds its representation through several stacked convolutional and pooling layers, composing local edges into increasingly abstract, multi-scale object parts; 2D-ROCKET, by construction, performs a single flat feature-extraction stage with no learned composition across layers (Sec. 2). The CNN baseline used for comparison in this paper is deliberately shallow (two convolutional layers, Sec. 3) to keep model complexity comparable across baselines, and consequently makes only limited use of the deep, multi-scale hierarchy that distinguishes modern CNN architectures such as ResNets – which, along with other deep or state-of-the-art models, are explicitly out of scope here. 2D-ROCKET should therefore not be expected to approach the accuracy of genuinely deep, multi-scale networks on datasets like CIFAR-10, where distinguishing object classes benefits from exactly this kind of hierarchical structure, even where it matches or slightly exceeds the shallow baseline tested here (Sec. 5). The method is instead proposed for texture-dominated or otherwise spatially repetitive image data – surface, material, or inspection imagery – where classification relies on local, statistically distinctive patterns rather than on composed object hierarchies, and where a flat random-feature bank can plausibly substitute for learned depth. CIFAR-10 is included primarily as a stress test against object-recognition structure that 2D-ROCKET cannot represent, while the texture-oriented case is examined directly in Sec. 5 via a proof-of-concept on KTH-TIPS and, at greater length and with a ten-seed statistical protocol, via three independent real-world datasets in Sec. 5.4.

This paper addresses three research questions: (i) how well does a ROCKET-style 2D feature-extraction approach perform on image classification compared to simple baselines, (ii) how do key design choices – kernel size, dilation, kernel count, and selected aggregation features – influence performance and computational cost, and (iii) what are the practical advantages, limitations, and application scenarios of the resulting 2D-ROCKET approach.

2. Related Work

Classical and lightweight approaches. Before the deep learning era, image classification commonly combined handcrafted feature descriptors with classical classifiers. Histogram of Oriented Gradients (HOG) summarizes gradient magnitude and direction in local cells (Dalal and Triggs, 2005), while Local Binary Patterns (LBP) encode local texture via thresholded neighborhoods (Pietikäinen, Hadid, Zhao and Ahonen, 2011). Both are typically paired with a Support Vector Machine (SVM) (Cortes and Vapnik, 1995), forming a well-understood, interpretable baseline. Lightweight convolutional networks, including shallow CNNs (SCNNs), bridge the gap between full deep networks and classical pipelines by using few convolutional layers and simple pooling while retaining learned, task-specific filters (Lei, Liu, Dai and Ling, 2019). CNNs remain the dominant approach for state-of-the-art image classification (Krichen, 2023), but their computational and memory cost motivates the search for lighter alternatives.

ROCKET and its variants. ROCKET extracts features from time series using a large ensemble (typically $\geq 10,000$) of randomly initialized, fixed convolutional kernels, summarized via the Proportion of Positive Values (PPV) and the maximum activation (Max), and classified with a linear Ridge classifier (Dempster et al., 2020a). The reliance on a large number of independent random kernels can be linked to the Johnson–Lindenstrauss lemma, which suggests that sufficiently many random projections approximately preserve pairwise distances (Ghojogh, Ghodsi, Karray and

Crowley, 2021). Follow-up work improved computational efficiency and determinism: MiniROCKET restricts kernel weights to two values and uses PPV only (Dempster, Schmidt and Webb, 2020b); MultiROCKET adds first-order differences and three further pooling statistics – Mean Positive Value (MPV), Mean Index of Positive Values (MIPV), and Local Share of Positive Values (LSPV) (Tan, Dempster, Bergmeir and Webb, 2022); Hydra introduces a dictionary-style kernel-competition mechanism (Dempster, Schmidt and Webb, 2022). This work adopts the MultiROCKET-style statistics (MPV, MIPV, LSPV) as candidate features for the 2D setting, since they are inexpensive and plausibly informative for spatially structured data.

Random convolutions on images. The idea of random projections and random convolutional features for images predates ROCKET: Rahimi and Recht map inputs into a randomized low-dimensional feature space that approximates a kernel, enabling fast linear learning on non-linear problems (Rahimi and Recht, 2007). Random depthwise convolutional architectures combined with global pooling and a linear classifier have achieved competitive accuracy on CIFAR-10 and STL-10 (Xue and Roshan, 2018), and random filters followed by pooling and a linear SVM have reached roughly 84% accuracy on facial-expression classification without any filter learning (Ren, 2016). Unlike these approaches, most of which retain a multi-layer, CNN-like architecture with stacked random convolutions, ROCKET performs a single, flat feature-extraction stage with simple statistical aggregation and no non-linear activations. To our knowledge, this flat ROCKET-style design has not been systematically adapted and ablated for 2D image data, which is the gap this work addresses.

3. Methods

3.1. 1D-ROCKET baseline

As a reference and sanity check, a minimal, independent 1D-ROCKET implementation follows Dempster et al. closely (Dempster et al., 2020a): images are flattened row-wise, kernels of length l are convolved with the flattened sequence x as

$$(x * k)[t] = \sum_{i=0}^{l-1} x[t+i] \cdot k[i], \quad (1)$$

and each kernel yields two features,

$$f_{\max} = \max_t (x * k)[t], \quad f_{\text{PPV}} = \frac{1}{L-l+1} \sum_t \mathbb{1}((x * k)[t] > 0). \quad (2)$$

Kernel sizes $k \in \{7, 9, 11\}$ and a per-kernel-group random choice between *same* and *valid* padding (deterministic given the seed) are used, followed by a Ridge classifier.

3.2. 2D-ROCKET design

2D-ROCKET replaces the flattening step with genuine 2D convolutions, preserving spatial (and, for RGB inputs, channel) structure. Kernel weights are sampled with variance $2/\text{fan-in}$ (He-style initialization) and a small random bias $\mathcal{N}(0, 0.1)$; stride is fixed to 1 to preserve full spatial resolution, which matters for location-sensitive statistics.

In addition to PPV and Max, three further statistics inspired by MultiROCKET (Tan et al., 2022) are evaluated: the Mean Positive Value (MPV, the average magnitude of positive activations), the Mean Index of Positive Values (MIPV, a first-order moment of the activation’s spatial location, computed per axis as $\text{MIPV}_y/\text{MIPV}_x$), and the Local Share of Positive Values (LSPV, the proportion of positive responses within predefined local regions). Eight feature-set variants (F1: PPV+Max, through F8: all five statistics) are compared in an ablation (Sec. 5).

Three padding strategies are compared: *same* (kernel applied at every position, image-size output), *valid* (kernel applied only where it fully fits, discarding kernel/dilation groups whose effective receptive field $(k-1) \cdot d + 1$ exceeds the image), and *random* (a per-kernel-group, seed-deterministic mix of the two).

Two normalization stages operate at different levels and are complementary: (i) dataset-level normalization (fixed per-dataset mean/std, applied by the loader) and (ii) an optional per-image, per-channel z-score standardization $x' = (x - \mu)/\max(\sigma, \epsilon)$ with $\epsilon = 10^{-6}$, applied immediately before convolution to make kernel responses scale-invariant. Extracted features are additionally standardized (zero mean, unit variance, fit on the training split) before being passed to a RidgeClassifier; the regularization strength α is selected from a grid $\{1000, 1200, \dots, 2400\}$ per experiment.

Table 1

Main results: test accuracy and classifier/training time per method and dataset. ROCKET times denote classifier fit time after (cached) feature extraction; CNN and SVM times denote full training time.

Method	Dataset	Accuracy	Time (s)
Minimal 1D-ROCKET	MNIST	0.9858	13.3
2D-ROCKET (final)	MNIST	0.9941	103.9
Lightweight CNN	MNIST	0.9906	77.8
Linear SVM	MNIST	0.9166	7816.6
Minimal 1D-ROCKET	CIFAR-10	0.6258	10.5
2D-ROCKET (final)	CIFAR-10	0.7343	90.2
Lightweight CNN	CIFAR-10	0.6962	79.3
Linear SVM	CIFAR-10	0.3790	6890.1

3.3. Baselines

A *linear SVM* is trained on flattened, standardized pixels (with a random projection to 2048 dimensions for CIFAR-10 to keep training tractable), $C = 0.1$, $\text{max_iter}=20,000$. A *lightweight CNN* consists of two convolutional layers (32 and 64 channels, ReLU, max-pooling) followed by a small MLP (128 hidden units), trained for 10 epochs with Adam ($\text{lr} = 10^{-3}$) and cross-entropy loss, without data augmentation or batch normalization.

4. Experimental Setup

Datasets. MNIST (60,000/10,000 train/test, 28×28 grayscale digits, normalized with mean 0.1307/std 0.3081) and CIFAR-10 (50,000/10,000 train/test, 32×32 RGB, 10 object classes, normalized with per-channel mean (0.4914, 0.4822, 0.4465)/std (0.2470, 0.2435, 0.2610)) serve as the two benchmark datasets.

Protocol and fairness. All experiments use a fixed seed (42) for Python, NumPy, and PyTorch, deterministic CuDNN settings, and identical train/test splits; no data augmentation, ensembling, or pretraining is used anywhere. Within each ablation, all variants share batch size, subsampling indices (when used), and classifier configuration, so that observed differences are attributable to the single factor under study. Accuracy on the held-out test split is the primary metric; classifier fit time (and, where noted, total runtime including feature extraction) is reported as a secondary, practically relevant metric.

Ablation strategy. Design decisions are propagated forward in stages: (1) feature-set selection (1D), (2) 1D vs. 2D kernels at matched budgets, (3) padding strategy, (4) preprocessing banks (raw vs. per-image-normalized channels), (5) kernel configuration (size/dilation) at a fixed kernel count, and (6) kernel-count scaling. Each stage’s winner defines the configuration carried into the next.

Environment. Experiments ran on a Google Colab high-memory runtime with an NVIDIA A100 GPU, Python 3.12, PyTorch 2.9, Torchvision 0.24, scikit-learn 1.6, and NumPy 2.0.

5. Results

5.1. Baselines

Table 1 summarizes the main comparison. The minimal 1D-ROCKET baseline already reaches 98.58% on MNIST but only 62.58% on CIFAR-10, reflecting that flattened 1D kernels discard the spatial structure that matters for natural images. The linear SVM trails clearly on both datasets (91.66% / 37.90%) and requires by far the longest training time (over two hours on either dataset), since it optimizes directly on high-dimensional pixel/projected features. The lightweight CNN reaches 99.06% / 69.62% with short, GPU-accelerated training.

5.2. Ablations

Feature set. Across eight combinations of PPV, Max, MPV, MIPV, and LSPV, accuracy increases monotonically with more statistics but with diminishing returns: on MNIST, PPV+Max alone reaches 98.58% (6,048 features), while the full set (F8, 15,120 features) reaches 99.02%. A reduced set (F5: PPV, Max, MPV, MIPV; 12,096 features) already attains 99.01% at roughly 75% of F8’s feature budget. On CIFAR-10 the pattern is similar but the absolute gain is

larger: PPV+Max reaches 62.58%, F5 reaches 65.68%, and the full set F8 reaches the best observed 65.92%. F5 is adopted for subsequent experiments as a favorable accuracy/cost trade-off.

1D vs. 2D kernels. At matched kernel budgets, 2D convolutions outperform their flattened 1D counterparts: 99.01%→99.14% on MNIST and, more notably, 65.68%→66.80% on CIFAR-10, at the cost of a larger feature dimension (15,120 vs. 12,096) and longer extraction/fit time. Subsequent experiments use 2D kernels throughout.

Padding. *Valid* padding gives the best CIFAR-10 accuracy (67.15%, vs. 66.99% for *same* and 66.79% for *random*) at equal feature dimensionality, and on MNIST it matches *same* padding (99.19% vs. 99.20%) while reducing the feature dimension from 15,120 to 13,385 and training time from 44.8 s to 37.1 s. Valid padding is adopted for the remainder of the study.

Preprocessing banks. Comparing raw (dataset-normalized) channels, per-image z-score-preprocessed channels, and their concatenation shows a dataset-dependent effect. On MNIST all three variants are within 0.2 percentage points of each other (best: gray-only, 99.19%). On CIFAR-10, per-image preprocessing *reduces* accuracy relative to raw RGB input (65.31% vs. 67.15%), and concatenating both banks performs worse still (62.33%), suggesting that per-image normalization removes inter-image color statistics that are informative for natural images. Raw (dataset-normalized) input is used going forward.

Kernel configuration and count. Among five kernel-size/dilation configurations at a fixed budget of 3,024 kernels, a mixed configuration with moderate dilation (sizes [3, 5, 7, 9], dilations [1, 2]) performs best on both datasets (99.34% on MNIST, 71.89% on CIFAR-10), outperforming both a single-scale undilated bank (the original ROCKET design, [7, 9, 11], dilation 1) and heavily dilated small kernels. Scaling this configuration's kernel count from 1,000 to 7,000 improves CIFAR-10 accuracy from 66.83% to 73.72%, with diminishing returns beyond roughly 5,000 kernels (73.42% at 5,000 vs. 73.72% at 7,000, at roughly double the training time). MNIST saturates even earlier, gaining less than half a percentage point from 3,000 to 5,000 kernels. $N = 5,000$ is selected as a practical compromise, tuned primarily for CIFAR-10.

5.3. Final configuration and per-class behavior

The final 2D-ROCKET pipeline (kernel sizes [3, 5, 7, 9], dilations [1, 2], $N = 5,000$ kernels, valid padding, features PPV+Max+MPV+MIPV, 25,000-dimensional feature vectors) reaches **99.41%** on MNIST and **73.43%** on CIFAR-10 (Table 1), a gain of 0.83 and 7.75 percentage points, respectively, over the corresponding 1D-ROCKET ablation baselines. On MNIST, per-class accuracy is uniformly high and slightly exceeds the CNN; on CIFAR-10, per-class accuracy is more variable across models, with the final 2D-ROCKET pipeline attaining the highest per-class accuracies among the tested methods but with larger spread than on MNIST, reflecting the higher intra-class variability of natural images.

As a proof of concept beyond MNIST/CIFAR-10, the same 2D-ROCKET configuration (with CIFAR-10-derived settings, unmodified) was evaluated on KTH-TIPS, a small ten-class texture dataset with varying lighting, viewpoint, and scale (Grauman and Taloen, 2004). 2D-ROCKET reached 94.7% accuracy, versus 43.5% for the linear SVM and 52.4% for the lightweight CNN baseline, with a classifier fit time around 0.66 s (versus 15.6 s for the SVM and 10.8 s for the CNN).

5.4. Real-world validation: industrial, medical, and texture imagery

The final 2D-ROCKET configuration (kernel sizes [3, 5, 7, 9], dilations [1, 2], $N = 5,000$ kernels, valid padding, features PPV+Max+MPV+MIPV_y+MIPV_x, 25,000-dimensional feature vectors), carried over unmodified as with KTH-TIPS, is evaluated on three real-world datasets: NEU-DET, a six-class hot-rolled steel strip surface-defect dataset (1,800 images), sharing its image source with the original NEU-CLS classification benchmark (Song and Yan, 2013) though introduced separately with (here unused) detection annotations; the Chest X-Ray Pneumonia dataset, 5,860 grayscale radiographs labeled NORMAL/PNEUMONIA (Kermany, Goldbaum, Cai, Valentim, Liang, Baxter, McKeown, Yang, Wu, Yan et al., 2018); and the Describable Textures Dataset (DTD), 5,640 images across 47 texture-attribute classes (Cimpoi, Maji, Kokkinos, Mohamed and Vedaldi, 2014). All three are accessed via their public Kaggle mirrors. Each dataset is resized to 128 × 128 grayscale and split 80/20 (stratified per seed, shared between methods), evaluated against the same lightweight CNN baseline (Sec. 3) trained for 30 epochs, longer than the 10 used for MNIST/CIFAR-10 and applied uniformly across all three datasets (Fig. 5 shows the resulting training-loss trajectory for DTD, the hardest of the three). Every (method, dataset) combination is run over ten seeds (42–51, reseeding data split, model initialization, and kernel sampling jointly), reported as mean ± population standard deviation with a paired *t*-test per dataset and a Holm–Bonferroni correction across the three tests (Table 2 reports both raw and adjusted *p*;

Table 2

Real-world validation: test accuracy and macro F1 (mean $\pm$ population std over 10 seeds), and total pipeline run time (single uncontended seed-42 run; data loading, training/feature extraction, evaluation) per method and dataset. All experiments use grayscale 128 $\times$ 128 inputs and a stratified 80/20 train/test split shared between both methods for a given seed. p -values are from a paired t -test (matched by seed, $n = 10$) on accuracy; p_{Holm} is the Holm–Bonferroni-adjusted value across the three tests in this table.

Method	Dataset	Accuracy	Macro F1	Time (s)
Lightweight CNN	NEU	0.925 ± 0.030	0.923 ± 0.031	57.1
2D-ROCKET	NEU	0.955 ± 0.012	0.955 ± 0.012	319.7
$p = 0.038$, $p_{\text{Holm}} = 0.038$ (ROCKET higher)				
Lightweight CNN	Chest X-Ray	0.958 ± 0.006	0.947 ± 0.008	197.1
2D-ROCKET	Chest X-Ray	0.935 ± 0.005	0.919 ± 0.006	1065.5
$p = 4.6 \times 10^{-7}$, $p_{\text{Holm}} = 9.2 \times 10^{-7}$ (CNN higher)				
Lightweight CNN	DTD	0.116 ± 0.016	0.112 ± 0.015	158.0
2D-ROCKET	DTD	0.282 ± 0.012	0.255 ± 0.013	1015.5
$p = 3.0 \times 10^{-10}$, $p_{\text{Holm}} = 9.0 \times 10^{-10}$ (ROCKET higher)				

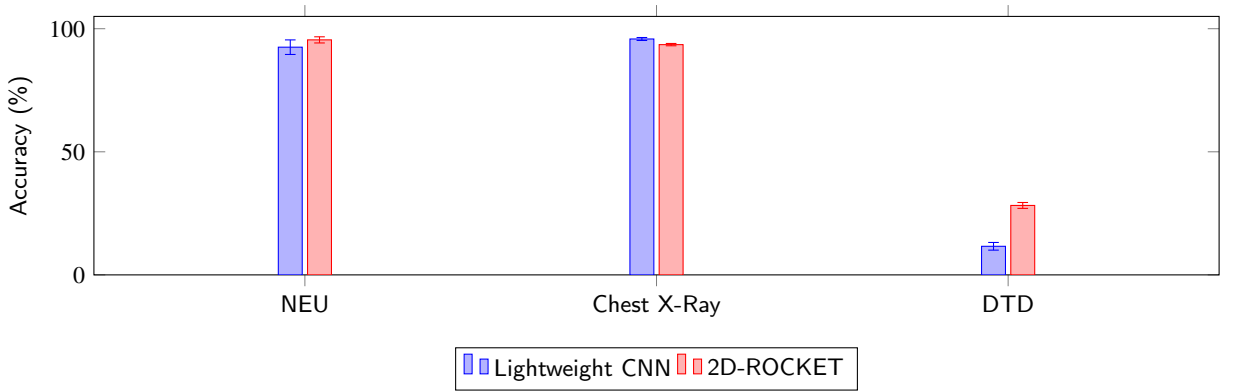


Figure 1: Test accuracy (% , mean $\pm$ std over 10 seeds) of the lightweight CNN baseline and the final 2D-ROCKET configuration on NEU, Chest X-Ray Pneumonia, and DTD. Differences are statistically significant on all three datasets (paired t -test, $n = 10$; Table 2).

all three remain significant at $\alpha = 0.05$, NEU’s adjusted value closest to the threshold). Reported times cover the full pipeline (data loading, training/feature extraction, evaluation) from a single uncontended seed-42 run, unlike Table 1’s cached-feature timings; the ten-seed sweep itself ran seeds concurrently on shared hardware, which inflates wall time and is not used for the time column. Secs. 5.4–5.6 ran on CPU (16-core/32-thread AMD Ryzen 9 9950X, 60 GB RAM, no GPU; Python 3.14, PyTorch 2.13, scikit-learn 1.9, NumPy 2.5), unlike the Colab/A100 environment used for Table 1 (Sec. 4) – run times across the two are not comparable.

Results are summarized in Table 2 and Fig. 1–5: 2D-ROCKET wins significantly on two of the three datasets and loses narrowly but significantly on the third.

On NEU, 2D-ROCKET reaches $95.5\% \pm 1.2$ against the CNN’s $92.5\% \pm 3.0$ ($p = 0.038$), reproducing the KTH-TIPS pattern with a smaller margin: the CNN’s range across seeds (85.8–96.7%) spans 2D-ROCKET’s entire range (92.2–96.7%). The source of the CNN’s higher seed sensitivity here (std 3.0 vs. 0.6 on Chest X-Ray) is not established – per-seed confusion matrices were not retained. Fig. 3 (seed 42) shows both models making few, evenly spread errors. Chest X-Ray Pneumonia is the one dataset where the CNN wins, $95.8\% \pm 0.6$ against $93.5\% \pm 0.5$ ($p = 4.6 \times 10^{-7}$), the most statistically confident result of the three given both methods’ low variance. Macro F1 (0.947 vs. 0.919) confirms the same ordering, so the result is not an artifact of the dataset’s class imbalance (855 PNEUMONIA vs. 317 NORMAL test images). Fig. 4 shows the gap concentrated in missed pneumonia cases: the CNN misses 23 of 855 (2.7%) against 2D-ROCKET’s 39 (4.6%).

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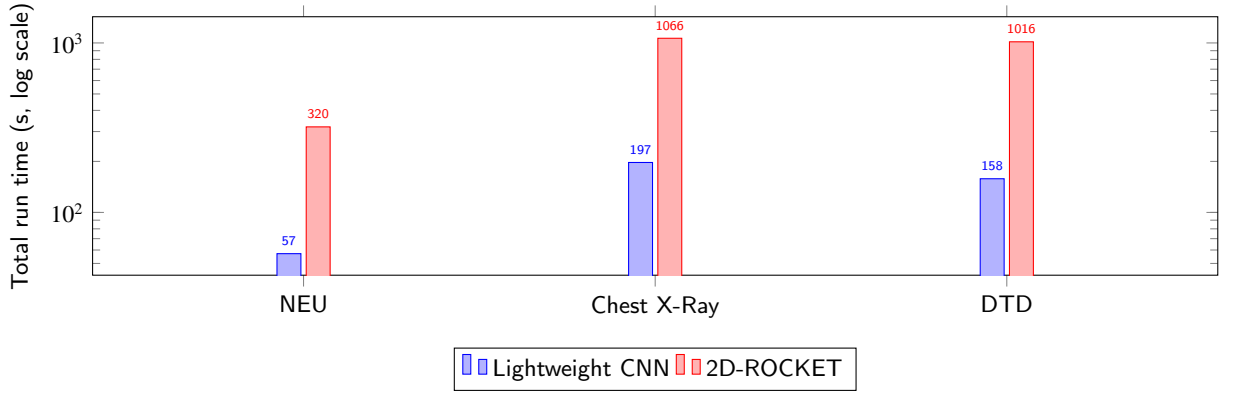


Figure 2: Total pipeline run time in seconds (log scale; data loading, training/feature extraction, and evaluation, uncached).

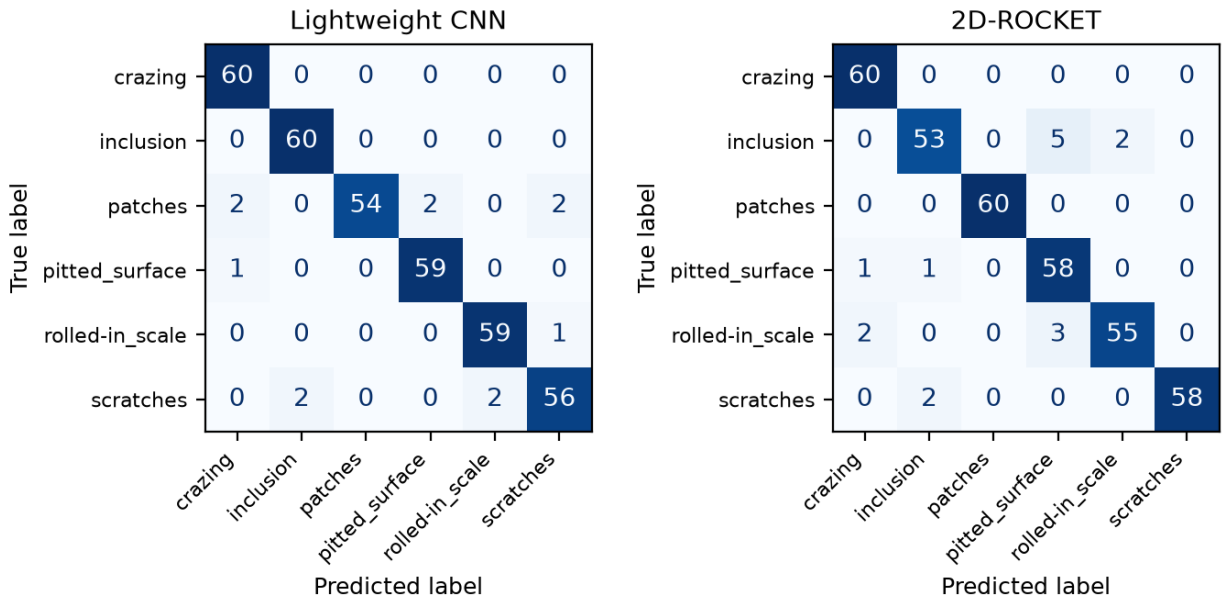


Figure 3: NEU confusion matrices for the representative seed-42 run (60 test images per class); see Table 2 for the full 10-seed accuracy distribution.

DTD gives the clearest result: 2D-ROCKET reaches $28.2\% \pm 1.2$ against the CNN's $11.6\% \pm 1.6$ ($p = 3.0 \times 10^{-10}$), a 16.6-point gap exceeding seed-to-seed noise on both sides. Fig. 5 (seed 42) shows the CNN memorizing its ~ 96 training images per class with no augmentation; that seed's 8.51% test accuracy is the lowest of the ten seeds run (mean 11.6%), so the memorization pattern is real but not necessarily its dominant driver of the ten-seed mean. 2D-ROCKET's ceiling (26.6–30.0% across seeds) sits far above the CNN's (8.5–13.8%) regardless of seed, consistent with a fixed-feature, linear-Ridge design being harder to overfit than a from-scratch CNN in a low-data, per-class regime. DTD also differs from NEU in class count (47 vs. 6), not only samples per class, and this design cannot separate the two as explanations for the CNN's weaker DTD performance; disentangling them needs a dataset varying class count and per-class samples independently, left for future work.

2D-ROCKET's uncached run time remains 5–6 $\times$ the CNN's across datasets (Table 2), since feature extraction is included here, unlike Table 1's cached-feature timings. These results reflect a single, CIFAR-10-tuned configuration with no dataset-specific retuning beyond the epoch-count adjustment above.

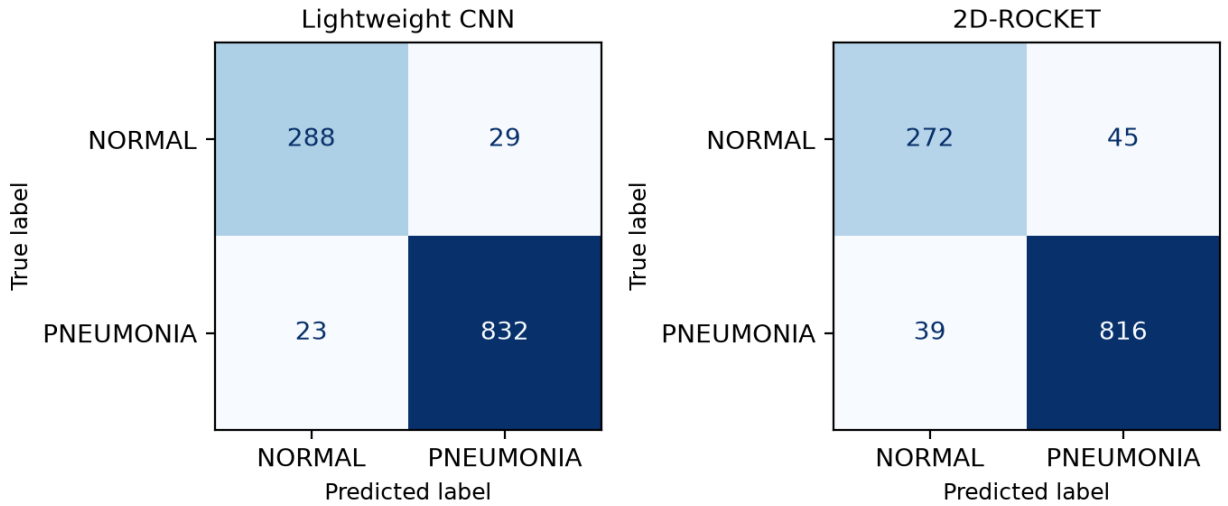


Figure 4: Chest X-Ray Pneumonia confusion matrices for the representative seed-42 run (317 NORMAL / 855 PNEUMONIA test images); see Table 2 for the full 10-seed accuracy distribution.

Table 3

Chest X-Ray Pneumonia: adding the LSPV feature (mean $\pm$ std over 10 seeds; capacity control over 5 seeds, 42–46). p -values are paired t -tests against 2D-ROCKET+LSPV, matched by seed ($n = 10$ for the first two rows, $n = 5$ for the capacity control).

Method	Accuracy	Macro F1	p vs. +LSPV
Lightweight CNN	0.958 ± 0.006	0.947 ± 0.008	0.22
2D-ROCKET	0.935 ± 0.005	0.919 ± 0.006	2.5×10^{-7}
2D-ROCKET + LSPV	0.956 ± 0.004	0.944 ± 0.006	–
2D-ROCKET, 8k kernels, no LSPV	0.938 ± 0.005	0.922 ± 0.006	2.4×10^{-3}

5.5. Closing the Chest X-Ray gap: restoring positional resolution

PPV, Max, and MPV pool over the entire activation map per kernel, and MIPV reduces spatial location to a single centroid, so a small, localized pneumonia infiltrate competes for influence against the same rib cage and mediastinal structure present in every image – a plausible reason a globally pooled feature bank struggles with localized pathology despite excelling on texture. MultiROCKET’s Local Share of Positive Values (LSPV, Sec. 2) was named as a candidate feature but not implemented elsewhere in this paper; it is added here as adaptive average-pooling of each kernel’s binary activation mask into a 3×3 grid of regional PPV values, giving each kernel coarse positional resolution at the cost of growing the feature dimension from 25,000 to 70,000.

Adding LSPV to Chest X-Ray (ten seeds, all else unchanged) gives $95.6\% \pm 0.4$ (macro F1 0.944 ± 0.006), against the CNN’s $95.8\% \pm 0.6$ and the original feature set’s $93.5\% \pm 0.5$ (Table 3). The gap to the CNN is no longer significant ($p = 0.22$), while the improvement over the original feature set is ($p = 2.5 \times 10^{-7}$). Mean run time (1,937 s) is essentially unchanged from 1,993 s without LSPV, since convolution over the kernel bank dominates cost far more than the Ridge fit or the added pooling step. 2D-ROCKET’s weakness on Chest X-Ray is therefore not an inherent property of random-kernel feature extraction but a specific, cheaply fixable consequence of discarding positional information during pooling.

LSPV also grows the feature dimension from 25,000 to 70,000, so part of its gain could be capacity rather than position. A capacity-matched control – 2D-ROCKET without LSPV but 8,000 kernels (40,000 features; memory constraints on the evaluation machine ruled out an exact 14,000-kernel match), five seeds (42–46) – reaches $93.8\% \pm 0.5$ (Table 3): significantly above the 5,000-kernel baseline on these seeds ($+0.55$ points, $p = 2.3 \times 10^{-4}$), but significantly

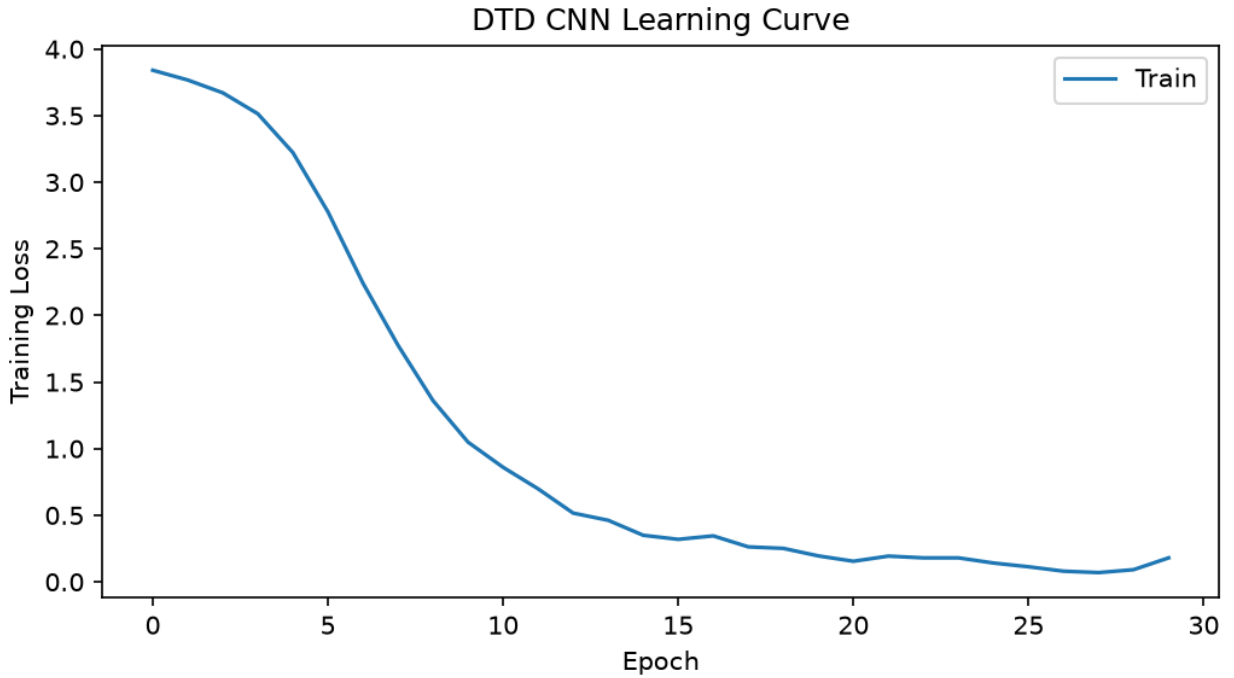


Figure 5: DTD lightweight-CNN training loss over the full 30-epoch schedule, seed 42. Loss falls below 0.2 with no plateau, consistent with memorizing the ~ 96 training images available per class (no augmentation). This seed’s test accuracy (8.51%) is the lowest of the ten seeds evaluated (mean $11.61\% \pm 1.56$, Table 2).

below LSPV (-1.50 points, $p = 2.4 \times 10^{-3}$). Capacity accounts for roughly a quarter of LSPV’s improvement (0.55 of 2.05 points); the remainder is attributable to positional resolution.

Scope is limited in two respects: LSPV was tested only on Chest X-Ray, not swept across NEU, DTD, MNIST, or CIFAR-10, where finer positional resolution could reduce the translation-invariance that plausibly helps texture classification; and the 3×3 grid was fixed a priori, not tuned against alternatives (e.g. 2×2 , 4×4). The result confirms that positional resolution closes the gap, not that this resolution is optimal.

5.6. SAR radio-frequency interference detection: a genuine time-series test

ROCKET was originally designed for genuine time series (Dempster et al., 2020a); everything in this paper so far has applied it to images (2D-ROCKET, and the flattened-1D-sequence baseline in Sec. 3). Raw Synthetic Aperture Radar (SAR) data is a natural genuine-time-series target: before image formation, a SAR instrument records one raw radar echo per transmitted pulse. Radio-frequency interference (RFI) detection in these echoes is an unsolved data-quality problem for missions such as Copernicus Sentinel-1, and, since labeled contamination events are comparatively rare, a genuinely small-data problem in practice. Given the sample-efficiency account developed for DTD (Sec. 6), this section tests whether that advantage extends to real, non-image, small-sample time series, using RFInject-v1-L0 (Del Prete et al., 2025), an ESA Φ -lab collection of Sentinel-1 Level-0 raw bursts with simulated RFI injected additively and stored separately from the clean baseline echo, giving exact clean/contaminated pairs of the same underlying signal.

One Sentinel-1 IW burst (102 azimuth lines, 19,654 complex range samples per line) is used; 11 of its 15 interference realizations contain non-zero injected energy (four are empty placeholders). Interference is sparse across azimuth lines as well as range samples – only 71 of 102 lines carry non-zero energy from at least one realization, and individual realizations typically affect only a handful of lines – so labels are assigned per (line, realization) pair with verified non-zero injected energy, 114 such pairs, rather than by an arbitrary line/realization pairing. Combined with the burst’s 102 clean lines, this gives 216 labeled examples.

Each line’s magnitude is block-max-pooled by a factor of 4 (preserving sparse interference peaks better than block-averaging) to 4,913 samples and per-series z-normalized. This 216-example, single-burst, unaugmented dataset is the low-data regime in which Sec. 6 predicts 2D-ROCKET should have a structural advantage over a from-scratch CNN.

Table 4

SAR RFI detection (clean vs. contaminated echo lines): mean $\pm$ std over 10 seeds, $n = 216$ examples total. p -values are paired t -tests ($n = 10$) against 1D-ROCKET.

Method	Accuracy	Macro F1	p vs. ROCKET
Linear SVM	0.768 ± 0.059	0.767 ± 0.061	0.026
1D-ROCKET	0.802 ± 0.038	0.802 ± 0.038	–
1D CNN	0.836 ± 0.040	0.836 ± 0.040	1.7×10^{-3}

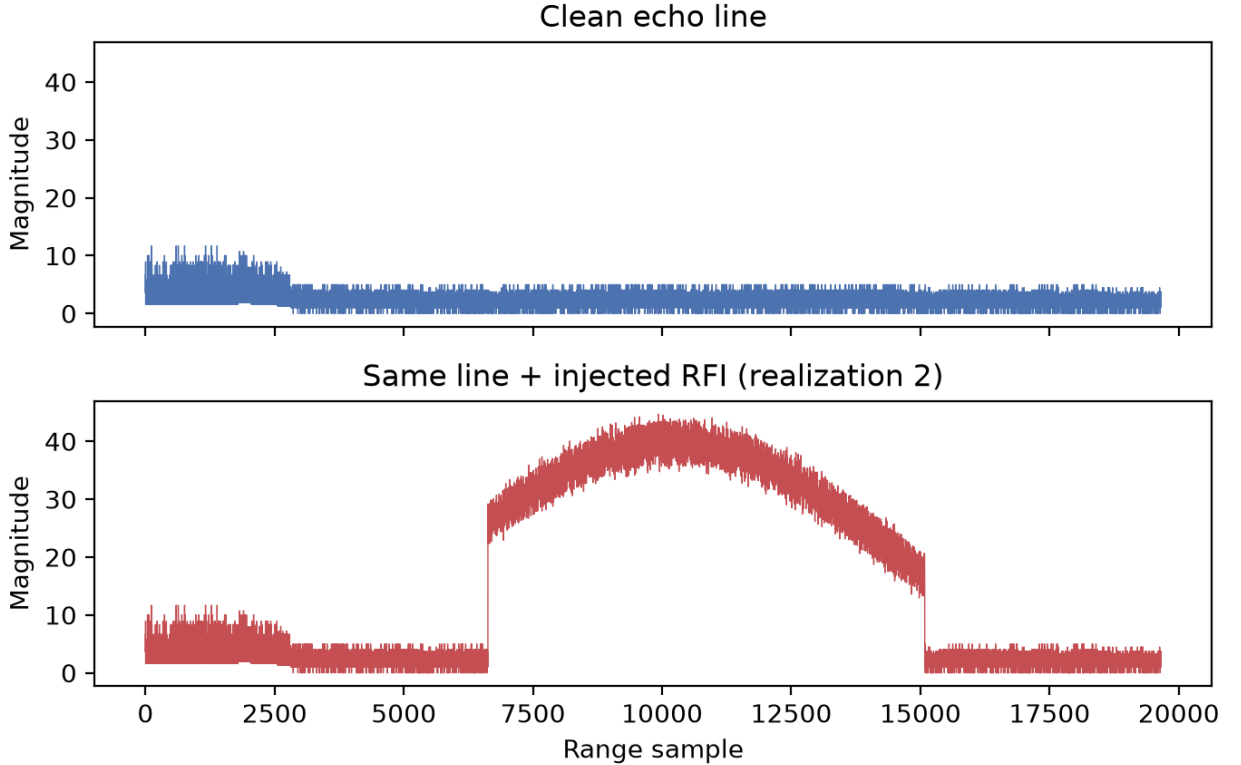


Figure 6: Magnitude of one raw Sentinel-1 echo line before and after additive RFI injection. The injected interferer raises the local energy envelope by roughly 4–5 $\times$ over a contiguous span of the line – a spatially localized anomaly, not a texture.

Three methods, freshly implemented for this section, are compared over 10 seeds (80/20 split reseeded jointly with model initialization, as in Sec. 5.4): a genuine 1D-ROCKET (kernel lengths $\{7, 9, 11\}$, dilations $\{1, 2, 4\}$, random per-kernel same/valid padding, PPV+Max, $N = 5,000$ kernels, RidgeClassifier, matching Dempster et al.’s original design), a shallow 1D CNN (two convolutional layers, 30 epochs), and a linear SVM ($C = 0.1$, matching this paper’s SVM methodology throughout).

1D-ROCKET reaches $80.2\% \pm 3.8$, significantly ahead of the linear SVM ($76.8\% \pm 5.9$, $p = 0.026$), consistent with the rest of this paper. It does not beat the CNN: the 1D CNN reaches $83.6\% \pm 4.0$ ($p = 1.7 \times 10^{-3}$ vs. 1D-ROCKET). Fig. 6 shows why: RFI here is a spatially localized anomaly – an elevated-energy span over part of the line – structurally the same problem as Chest X-Ray’s localized opacity (Sec. 5.4), not a texture repeated across the signal. A CNN’s position-adaptive filters suit this regardless of sample count; 1D-ROCKET’s PPV and Max still summarize each kernel’s response as one global number per series. This refines rather than confirms the DTD-derived hypothesis: low data alone does not predict a ROCKET advantage – it is low data combined with a texture-like, not localized-anomaly, target.

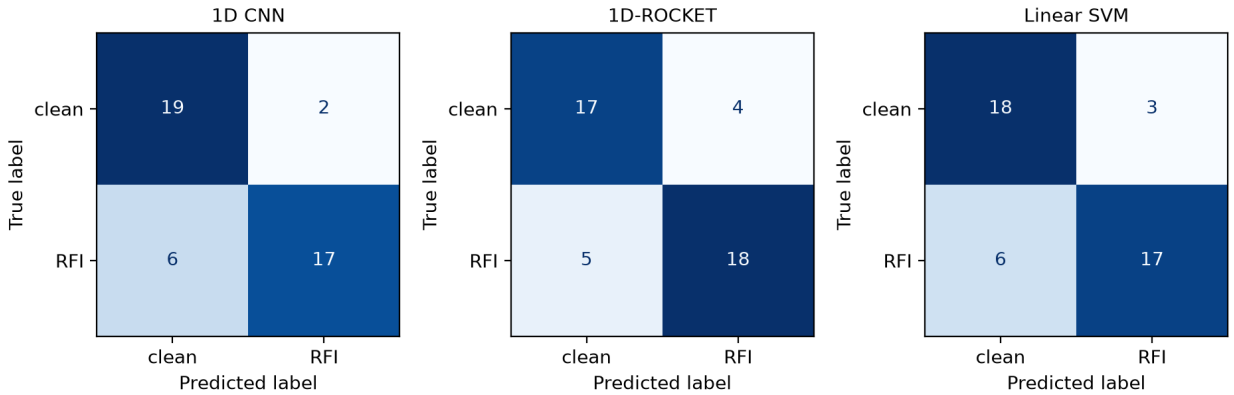


Figure 7: SAR RFI detection confusion matrices for the representative seed-42 split (22 clean / 22 RFI test examples).

This is an initial proof of concept, not a definitive result: it rests on a single burst and injection source, with no claim about generalization across bursts, sensors, or real interference; the $4\times$ block-max downsampling factor was not ablated against alternatives. As in Sec. 5.5, adding LSPV-style regional pooling to 1D-ROCKET is a natural untested next step for narrowing the CNN gap.

6. Discussion

The ablations indicate that most of 2D-ROCKET’s discriminative power comes from a small, targeted set of local statistics: PPV and Max alone already capture the bulk of the signal, and additional statistics yield incremental gains at a clear feature-dimension and runtime cost. Preserving spatial structure via genuine 2D convolutions, rather than flattening images into 1D sequences, is the single largest architectural lever, particularly on CIFAR-10 where spatial and color structure is more informative than on MNIST’s simple binary-like strokes. Kernel *configuration* (mixed sizes with moderate dilation) matters more than raw kernel *count*: multi-scale coverage consistently beats single-scale or heavily dilated banks, and accuracy plateaus once the kernel budget exceeds roughly 5,000, after which feature dimensionality and training time keep growing with little additional benefit. Padding behaves as a cheap, implementation-level regularizer: restrictive (valid) padding discards many boundary/large-receptive-field features that turn out to be largely redundant for these datasets, reducing cost without hurting accuracy. Preprocessing choices, by contrast, are dataset-dependent – per-image normalization is neutral on grayscale digits but actively harmful on natural color images, since it removes inter-image color cues that carry class information.

Relative to the baselines, 2D-ROCKET slightly outperforms the lightweight CNN on both datasets (by 0.35 points on MNIST, 3.81 points on CIFAR-10) while requiring no gradient-based training of its feature extractor. This result has to be read in light of what the CNN baseline actually is: two convolutional layers, no batch normalization, no augmentation, chosen deliberately to keep its complexity comparable to the other lightweight baselines (Sec. 3). Such a shallow network itself composes features across only two layers and therefore exercises little of the deep, multi-scale hierarchical composition – stacking many layers so that later ones respond to combinations of earlier, smaller-scale features – that characterizes deeper CNN architectures (e.g., ResNets), which were explicitly excluded from this study’s scope. Beating this particular baseline is consequently not evidence that 2D-ROCKET’s flat, single-stage design can match deep, multi-scale CNNs in general: 2D-ROCKET has no mechanism at all for combining kernel responses across stages into higher-order, compositional features, whereas even a moderately deep CNN does. On object-recognition tasks where classes are distinguished by exactly this kind of multi-scale, compositional structure, a genuinely deep CNN would be expected to open a gap that no amount of additional random 2D-ROCKET kernels can close, since kernel count and diversity scale the width of a single stage, not its depth. 2D-ROCKET does, however, decisively outperform the linear SVM in both accuracy and training time on both datasets (the SVM takes roughly two hours per dataset, dominated by kernel-free but high-dimensional optimization, versus under two minutes for 2D-ROCKET’s classifier fit). This positions 2D-ROCKET as an attractive middle ground for the right kind of data: more expressive than classical

handcrafted-feature pipelines, cheaper to train and more transparent than a CNN, but only where classification does not hinge on hierarchical, multi-scale composition.

The strongest practical case for the approach is suggested by the KTH-TIPS result: on a texture-dominated dataset, where local, repetitive statistics rather than multi-scale object composition drive classification, 2D-ROCKET's advantage over both baselines is far larger than on MNIST/CIFAR-10 – including over the CNN, which gains comparatively little from its hierarchical structure when there is no object hierarchy to compose. This is consistent with the intuition that random local filters combined with simple statistics are naturally well matched to repetitive, texture-like structure – the kind found in industrial surface inspection, material classification, or other structured imaging domains – even without any domain-specific tuning, and it is this regime, not general object recognition, that the method is proposed for.

The ten-seed real-world evaluation (Sec. 5.4) refines this paper's texture-domain narrative: 2D-ROCKET wins significantly on NEU and DTD and loses significantly, by a smaller margin, on Chest X-Ray Pneumonia. Two factors appear to drive this, compounding on DTD: (i) *texture-likeness* – NEU and DTD favor 2D-ROCKET's flat statistics, while Chest X-Ray's localized opacities favor the CNN's spatially adaptive filters; and (ii) *sample scarcity without augmentation* – DTD's ~ 96 images/class push the from-scratch CNN toward memorization (Fig. 5), while 2D-ROCKET's linear Ridge fit over fixed features is harder to overfit. NEU's ~ 240 images/class is not a low-data regime in this sense (CNN mean 92.5%), so its win is attributable to factor (i) alone, while DTD's larger margin (16.6 vs. NEU's 3.0 points) reflects both factors together. This predicts that restoring positional resolution should narrow the Chest X-Ray gap specifically. Sec. 5.5 confirms it: adding LSPV drops the CNN's advantage from significant ($p = 4.6 \times 10^{-7}$) to indistinguishable ($p = 0.22$), and a capacity-matched control attributes only about a quarter of that gain to added feature count, the rest to positional resolution.

Sec. 5.6 provides a third, independent test in a different data modality (raw 1D SAR time series), chosen to probe factor (ii) in isolation: with only 216 examples, sample scarcity alone would predict a 1D-ROCKET win. It does not win – a CNN wins there too ($p = 1.7 \times 10^{-3}$) – because the target is a localized energy anomaly (Fig. 6), the same category as Chest X-Ray's opacities, not a texture. Low data alone is therefore not sufficient for a ROCKET-family advantage without factor (i) also holding. Three of the four factor combinations have now been observed directly – texture/adequate-data (NEU), texture/scarse-data (DTD), non-texture/scarse-data (SAR) – with non-texture/adequate-data (Chest X-Ray) as the fourth, a small but genuine 2×2 probe of the hypothesis.

Limitations: single-seed evaluation for the MNIST/CIFAR-10/KTH-TIPS ablation chain (small effect sizes, especially on near-saturated MNIST, warrant caution without replication); a Ridge regularization strength ($\alpha = 1200$) tuned once on MNIST/CIFAR-10-scale data and never re-tuned for the smaller real-world training sets or the 70,000-dimensional LSPV feature space; deliberately simple baselines throughout; and growing memory/conditioning pressure as feature dimensionality reaches the tens of thousands. The paired t -tests in Sec. 5.4 assume rather than verify normality at $n = 10$, and since the train/test split is reseeded jointly with model initialization, seed-to-seed variance conflates model-fitting and test-set-composition variability.

7. Conclusion

This work adapted ROCKET's random-kernel, statistics-based paradigm from one-dimensional time series to two-dimensional image classification. Preserving spatial structure through genuine 2D convolutions, combined with a compact set of aggregation statistics (PPV, Max, MPV, MIPV) and a multi-scale, moderately dilated kernel bank, is the key design choice that makes 2D-ROCKET effective within a single flat feature-extraction stage. The resulting pipeline reaches 99.41% accuracy on MNIST and 73.43% on CIFAR-10 without any learned convolutional weights, and slightly outperforms a deliberately shallow, two-layer CNN baseline on both datasets, in addition to clearly outperforming a linear SVM in accuracy and training time. This should not be over-interpreted: the shallow CNN baseline itself makes limited use of the deep, layered composition that lets genuinely deep networks assemble multi-scale object representations, and such deeper architectures were out of scope for this comparison. Given its flat, single-stage architecture, 2D-ROCKET has no mechanism to close that kind of gap regardless of kernel count, and this paper does *not* propose it as a competitor to deep CNNs on natural-object image classification. A proof-of-concept on the texture-dominated KTH-TIPS dataset (94.7% vs. 43.5%/52.4% for SVM/CNN) instead points to the domain where the approach is intended and most effective: repetitive, texture-like image data – material and surface textures, inspection imagery, and similar structured signals – where local statistics rather than composed, multi-scale object structure drive classification, and where the architectural advantage of deep networks over a flat random-feature bank is far smaller.

2D-ROCKET is best understood as a lightweight, reproducible, interpretable baseline for texture-dominated and/or low-data, low-augmentation regimes, not a replacement for deep networks on general vision tasks requiring multi-scale composition. A ten-seed real-world evaluation (Sec. 5.4) applied the unmodified, CIFAR-10-tuned configuration to industrial surface-defect (NEU), texture-attribute (DTD), and medical (Chest X-Ray) imagery, winning significantly ($p < 0.05$) on two of three datasets and losing narrowly, significantly, on the non-texture Chest X-Ray task. The size of 2D-ROCKET's advantage tracked both texture-likeness and per-class sample scarcity, with DTD's combination of both producing the largest margin (16.6 points, $p = 3 \times 10^{-10}$). The texture-likeness explanation was tested directly: adding LSPV (Sec. 5.5) closed the Chest X-Ray gap to indistinguishability ($p = 0.22$) at no runtime cost, with a capacity-matched control attributing three-quarters of that gain to positional resolution rather than added capacity. A further test on genuine 1D time series – raw Sentinel-1 SAR echo lines, a freshly implemented 1D-ROCKET, a deliberately small (216-example) RFI-detection task (Sec. 5.6) – significantly beat a linear SVM but lost to a CNN, since the target there is a localized anomaly rather than a texture: sample scarcity alone does not predict a ROCKET-family advantage without texture-likeness also holding. Promising directions for future work include multi-seed validation for the MNIST/CIFAR-10/KTH-TIPS ablation chain, sweeping LSPV across the texture-dominated datasets, domain-aware kernel sampling, hybrid pipelines recovering some hierarchical composition, dimensionality reduction at large kernel budgets, and varying training-set size and augmentation independently of texture content to further disentangle the two factors identified in Sec. 6.

Data and Code Availability

Code for the real-world validation study (Sec. 5.4), the LSPV feature and capacity-control experiment (Sec. 5.5), and the SAR RFI detection study (Sec. 5.6), including scripts producing Tables 2–4 and Figs. 1–7, is available in the accompanying repository (`docs/REPRODUCING.md`). NEU, Chest X-Ray Pneumonia, and DTD are third-party datasets accessed via public Kaggle mirrors and not redistributed; the SAR burst (Sec. 5.6) is from the public RFIInject-v1-L0 Hugging Face bucket (Del Prete et al., 2025), likewise not redistributed. The MNIST/CIFAR-10/KTH-TIPS ablation study (Table 1 and the sections preceding Sec. 5.4) predates and is independent of this repository and is not reproducible from the released code; it is documented in the corresponding bachelor thesis.

CRedit authorship contribution statement

Felicitas Lock: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Christopher Bonenberger:** Supervision, Writing – review & editing.

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